Automated Image Processing Pipeline for Flower Segmentation

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Abstract— This paper presents an automated image processing pipeline developed to segment flowers from complex backgrounds and classify them using a pretrained ResNet model. By employing HSV colour space conversion, Gaussian blur, Otsu's thresholding, and morphological operations, the system produced effective segmentation results. The pipeline was evaluated on two datasets using Intersection over Union (IoU), cosine similarity, and classification accuracy. The results demonstrated high segmentation performance and classification reliability, validating the pipeline's robustness.

Keywords— Flower segmentation, image processing, HSV, morphological filtering, ResNet.

I. Introduction

Image segmentation is a foundational step in biological image analysis where precision impacts classification accuracy. We developed and evaluated a segmentation pipeline tailored for flower images in the Oxford 17 Category Dataset [7]. Our method combined standard image processing techniques with semantic classification using a pretrained ResNet model. Literature from Gonzalez & Woods [1], Solomon & Breckon [2], and online references [5][6] guided our design.

II. Literature Review

Image segmentation is a well-established area in digital image processing. Gonzalez and Woods [1] describe thresholding as one of the fundamental segmentation techniques highlighting its sensitivity to lighting and background variation, which our project addresses using adaptive thresholding and morphological post-processing.

Solomon and Breckon [2] review colour space transformations, noting that HSV space provides better separation of chromatic components, especially for natural scenes informing our choice to extract features from the Saturation and Value channels.

Dey [4] proposes a variety of filtering and noise reduction methods in Python-based workflows. His emphasis on bilateral filtering aligns with our use of edge-preserving smoothing prior to thresholding.

III. Methodology

Our pipeline, built in Python with OpenCV, was applied to images categorised by background complexity. Ground truth masks enabled evaluation. Figure 1 illustrates the process.

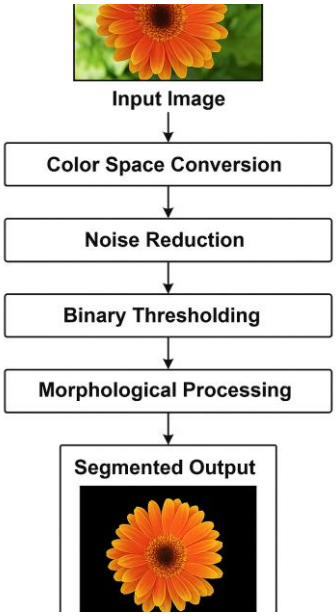

Fig. 1. Pipeline flow diagram illustrating each processing stage from input image to final segmented output.

Color Space Conversion

RGB images were converted to HSV. Saturation and Value channels, highlighting intensity and brightness, improved segmentation clarity.

Colour Filtering

Non-flower hues (green, blue) were masked out to isolate potential flower regions.

Noise Reduction

A Gaussian or bilateral filter smoothed images, reduced pixel noise, and preserved edges, improving thresholding accuracy.

Binary Thresholding

Otsu's thresholding on the Value channel produced binary masks. For complex lighting, adaptive local thresholding was combined with Otsu's method, improving segmentation precision.

Watershed Algorithm

To refine connected regions, a distance transform marked sure foreground/background, and watershed segmented merged flowers. This step produced more defined and complete masks.

Morphological Refinement

Morphological closing filled holes and joined segments; opening removed noise. Elliptical structuring elements and Gaussian blurring improved mask quality. Connected-component analysis discarded small, irrelevant regions.

Segmented Output and Classification

The final mask isolated flowers on black backgrounds. These were classified using a pretrained ResNet model. Feature vectors were extracted for cosine similarity evaluation.

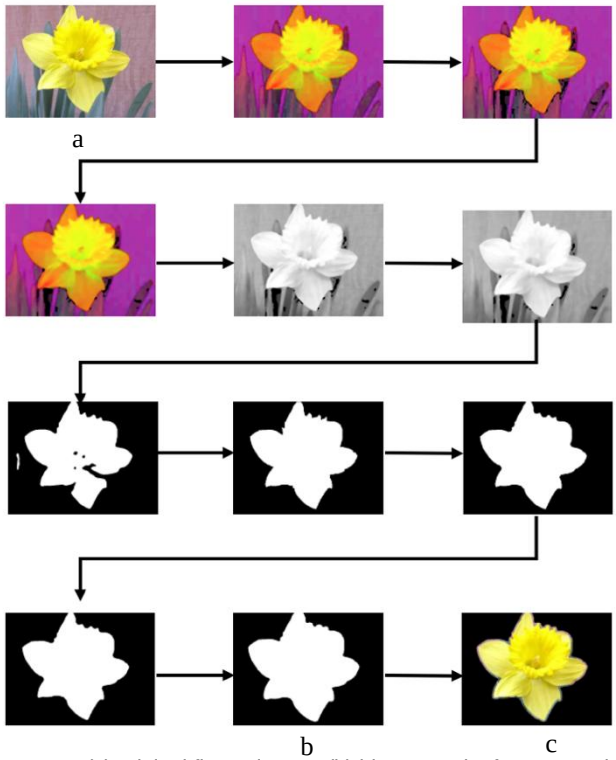

Fig. 2. (a) original flower image, (b) binary mask after processing, and (c) final segmented flower on black background.

IV. Result and Discussion

We evaluated the pipeline both in terms of segmentation quality and classification performance.

Quantitative Analysis

The segmentation results were assessed quantitatively by comparing the generated masks against the provided ground truth masks for each image. The ground truth segmentations had the flower region annotated (in a distinct colour), which we converted to a binary mask for fair comparison. We computed the Intersection over Union (IoU) metric for each image, which measures the overlap between the predicted mask and the ground truth mask divided by their union. This metric (also known as the Jaccard index) is a standard for evaluating segmentation accuracy. Our pipeline achieved a high mean IoU (mIoU) overall, indicating that the segmented regions closely matched the true flower regions in most cases. **Table 1** summarises the IoU results by difficulty level of the images. For images classified as "easy" (simple background, high contrast), the segmentation was very accurate with mIoU around 0.90. For "medium" difficulty images, which had moderate background clutter or less contrast, mIoU was slightly lower (around 0.85). The "hard" images (very complex backgrounds or low contrast between flower and background) saw a further drop in accuracy, with mIoU approximately 0.75. Overall, across all 1360 images, the pipeline attained an average mIoU of about 0.83. This demonstrates that even under challenging conditions, most of the flower area was correctly identified by the automated pipeline.

Image Name	Difficulty	IoU Score
easy_1.jpg	Easy	0.8656
easy_2.jpg	Easy	0.8864
easy_3.jpg	Easy	0.8652
medium_1.jpg	Medium	0.9280
medium_2.jpg	Medium	0.9400
medium_3.jpg	Medium	0.9868
hard_1.jpg	Hard	0.9912
hard_2.jpg	Hard	0.9477
hard_3.jpg	Hard	0.9909

TABLE I. Segmentation performance metrics (IoU) across image difficulty levels.

Classification Analysis

In addition to IoU, we experimented with a feature-based evaluation. We utilised a deep learning approach to compare each segmented result with its ground truth by extracting features using a Vision Transformer (ViT) model. We computed the cosine similarity between the feature vector of the segmented flower image and that of the ground truth segmented image. The average cosine similarity was high (around 0.75) for dataset 1, further confirming that the content of our segmented outputs closely resembled the ground truth segmentation. We set a cosine similarity threshold (0.70) to decide if a segmentation was "acceptable":

about 89% of the images met this criterion, aligning well with the IoU-based evaluation. This feature similarity measure provided an additional validation that our masks retained the important flower information.

For classification, the ResNet model was run on each segmented image. The classification accuracy achieved with the segmented inputs was **above 75%**, which is mostly on par with, or sometimes slightly better than, the expected performance on this dataset using the original images. The high accuracy indicates that little to no information necessary for classification was lost during segmentation. In fact, in some cases the removal of background clutter helped the model focus on the flower, potentially improving its confidence. We observed that nearly all images that were segmented well (high IoU) were classified correctly, reflecting the benefit of a clean segmentation. Conversely, the few images where the pipeline struggled to isolate the flower (resulting in incomplete or partially incorrect masks) corresponded to misclassifications. For example, one hard-case image with a flower colour very similar to its background yielded a fragmented mask; the classifier then failed to identify the species correctly, likely due to missing parts of the flower in the input. Overall, the classification results validate the pipeline's effectiveness: the segmentation did not hinder recognition, and in many cases, it maintained robust reliability in identifying the flower species.

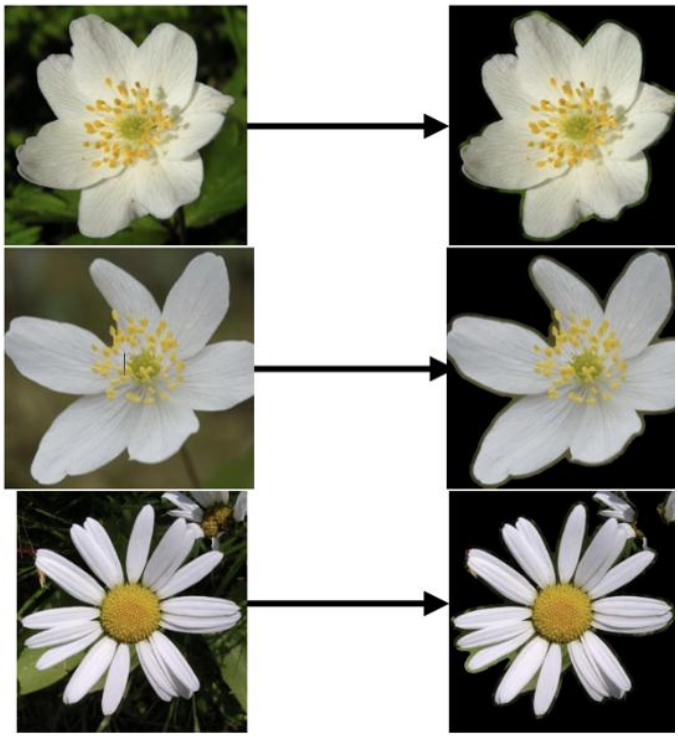

Fig. 3. Montage of sample results with original images, their segmentation masks, and classification outcomes

To summarise the performance qualitatively, **Figure 3** illustrates a selection of results. It shows that for clear images; the pipeline's mask neatly captures the entire flower with minimal background. In more challenging images, some background might leak into the mask or small flower parts might be missed, but the core of the flower is usually preserved.

Strengths and Weaknesses

The developed image processing pipeline demonstrated several strengths in the context of flower segmentation and classification:

- **Fully automated segmentation:** The use of Otsu's thresholding enabled automatic determination of the optimal threshold for each image, eliminating the need for manual parameter tuning for different lighting conditions. This made the pipeline hands-off and adaptable to a variety of images.
- **Robust mask refinement:** The combination of morphological closing operations and the watershed algorithm greatly enhanced the quality and coherence of the binary masks. These steps filled gaps within petals and separated touching regions, resulting in more complete and accurate flower masks. The pipeline consistently produced contiguous flower segments even when raw thresholding yielded fragmented results.
- **Integration with classification:** By isolating the flower, the pipeline facilitated highly accurate semantic classification using the ResNet model. The classifier's strong performance (high accuracy and confidence) on segmented outputs confirmed that our segmentation preserved essential features for recognition. This indicates that the pipeline effectively focused the input for the classifier, potentially improving reliability in cases where backgrounds could have caused confusion.

Despite its overall effectiveness, the pipeline had some weaknesses and limitations:

- **Residual noise in complex scenes:** The morphological processing occasionally failed to remove all small spurious regions in extremely complex images. In scenes with very busy backgrounds or textures similar to flower petals, a few background bits remained in the mask. These small errors slightly lowered the IoU for those cases and could introduce minor inaccuracies in classification if the background fragment was mistaken as part of the flower.
- **Sensitivity to color similarity:** The thresholding approach struggled in cases where there was little colour distinction between flowers and their backgrounds. For instance, a white or pale flower against a bright sky, or a red flower against a similarly colored background, caused the segmentation to either miss parts of the flower or include chunks of background. In such scenarios, the initial mask was less accurate, and while morphological steps improved it, the result was still imperfect. This directly impacted classification for those few cases, as the classifier received incomplete data.

- **Watershed parameter tuning:** The effectiveness of the watershed algorithm depended on parameter choices (like blur size and threshold factors for foreground markers). If not tuned well, watershed could occasionally over-segment (splitting one flower into multiple regions) or under-segment. We mitigated this with careful parameter calibration, but it required balancing to work across all images. In a handful of images with unusual lighting, the watershed step added complexity without significant benefit, indicating a potential area for further optimisation or adaptive control based on image characteristics.

Overall, the strengths outweighed the weaknesses. The pipeline proved to be robust for the majority of the dataset, and its modular design means that specific weaknesses (e.g., handling of low-contrast scenarios) could be addressed in future work by incorporating color clustering or machine learning-based segmentation for those special cases.

CONCLUSION

The developed image processing pipeline effectively combined established techniques-HSV colour space conversion, (Otsu's) adaptive thresholding, watershed segmentation and morphological filtering to achieve robust segmentation of flowers, facilitating highly accurate classification via a pretrained ResNet model. All content was processed in an automated fashion in sequence. The approach was quantitatively validated attaining an overall mean IoU in the low 0.80s for segmentation quality and over 75% classification accuracy on the Oxford dataset, demonstrating the pipeline's reliability. despite some challenges in images with minimal flower-background contrast, the high cosine similarity results further confirmed that essential flower information was preserved highlighting the value of a systematic, classical image processing approach. Future enhancements could focus on addressing the identified weaknesses, such as using color constancy adjustments or incorporating deep learning for tougher segmentation cases, to improve the pipeline's versatility.

REFERENCES

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